Screen time, sleep and adolescent depression

What should schools and health organizations do?

German Vargas | Project 3 | Data: 4,810 adolescents (Hökby et al., 2025)

We asked: do screens or sleep drive adolescent depression?

BUSINESS QUESTION

How do screen time and sleep quality affect adolescent depression risk, and what should health and educational organizations do to prevent it?

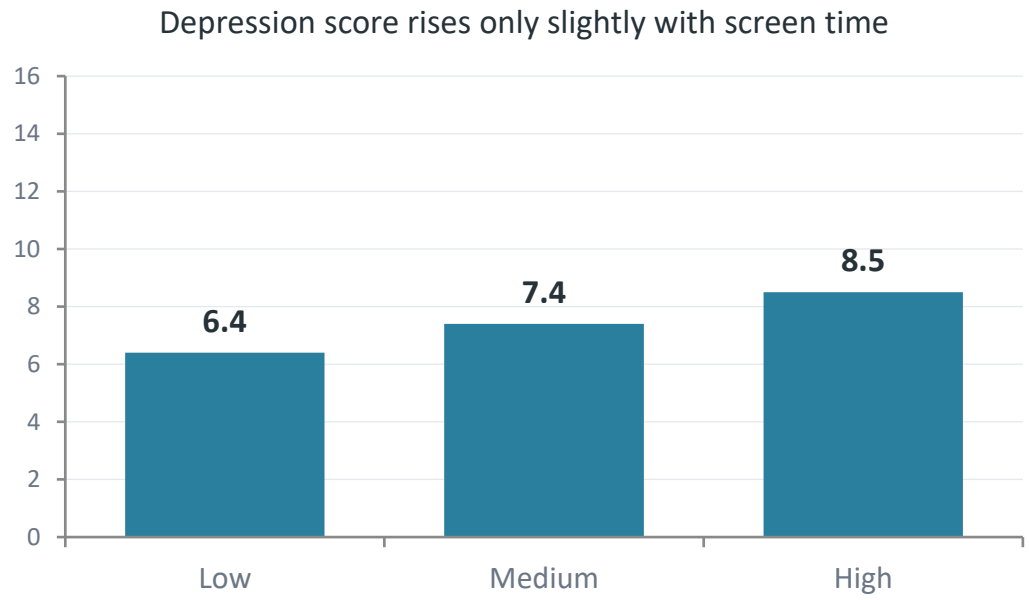
4,810 adolescents followed over 12 months

0-63 depression score (BDI-II) as the outcome

16.4% score in the clinical range (14 or higher)

Data: Hökby et al. (2025), PLOS Global Public Health (public dataset on Kaggle).

Screen time alone tells us little



SCREEN TIME EXPLAINS

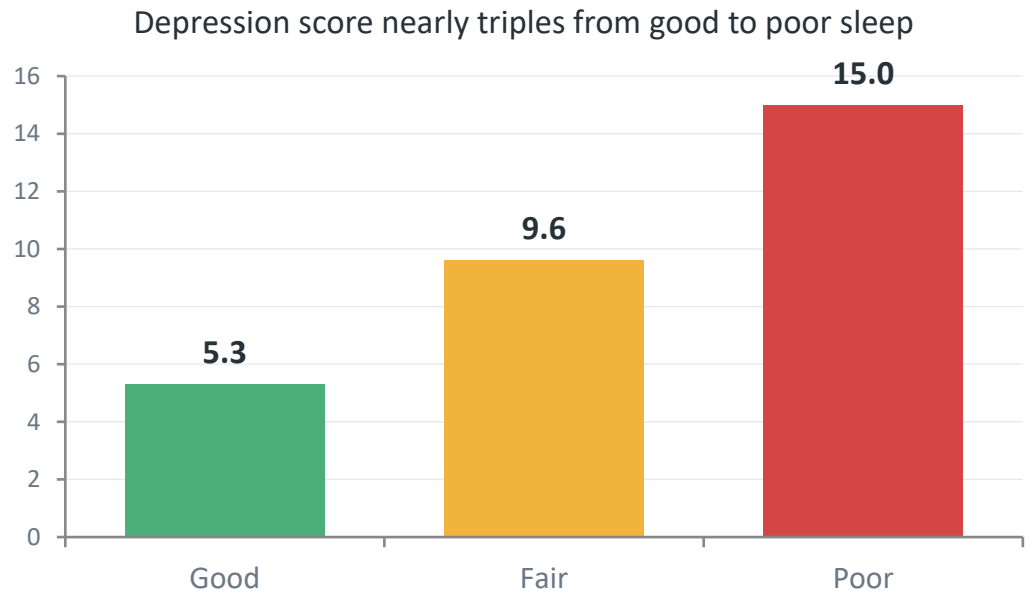
1%

of the differences in depression scores

Even the highest-screen group (8.5) stays below the clinical threshold of 14.

Mean BDI-II score by daily leisure screen time group (Low, Medium, High). n = 4,810.

Sleep quality tells us much more



14%

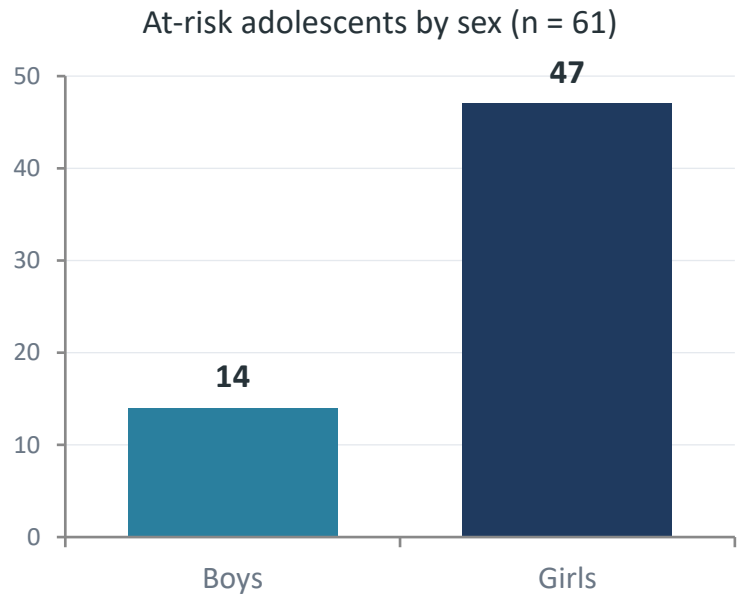
explained by sleep quality, 13x more than screen time

50% vs 9%

of poor vs good sleepers are classified as depressed

Mean BDI-II score by sleep quality group (Good, Fair, Poor). Same scale as the previous slide. n = 4,810.

High screen time plus poor sleep flags a high-risk group



77%

of the at-risk group are girls

16.3 vs 7.2

average depression score, at-risk vs everyone else (2.3x)

6.2 vs 7.7 h

sleep per night, at-risk vs everyone else

At risk = high screen time and poor sleep (61 of 4,810 adolescents).

What we recommend

Screen sleep quality at school check-ups

It explains 14% of depression scores, 13x more than screen time.

Focus on falling asleep and wake times

The strongest sleep links are difficulty falling asleep (r = 0.37) and early waking (r = 0.32).

Use high screen time plus poor sleep as a triage flag

It finds a group with 2.3x higher depression scores.

Prioritize girls and keep watching boys

77% of the at-risk group are girls; boys may be under-detected.

Bottom line: protect sleep first

Screen time affects depression mostly through sleep. The most effective programs protect sleep quality, especially falling asleep and consistent wake times.

PROPOSED NEXT STEPS

- Pilot sleep screening in a few schools
- Track sleep and depression scores over time
- Use the interactive dashboard to monitor at-risk groups